



Booth's Algorithm

```
1. import java.util.Scanner;
2. public class Booth {
3.     public int multiply(int n1, int n2)
4.     {
5.         int m = n1;
6.         int r = n2;
7.         int A = n1;
8.         int S = -n1;
9.         int P = 0;
10.        int count = Integer.SIZE;
11.        System.out.print(count);
12.        while (count > 0) {
13.            if ((r & 1) == 1) {
14.                P += A;
15.                S += m;
16.            }
17.            A <<= 1;
18.            S <<= 1;
19.            count--;
20.            r >>= 1;
21.        }
22.        return P;
23.    }
24. }
```

```
1. public static void main(String[] args) {
2.     Scanner scan = new Scanner(System.in);
3.     Booth b = new Booth();
4.     System.out.println("Enter two integer numbers -");
5.     int n1 = scan.nextInt();
6.     int n2 = scan.nextInt();
7.     int result = b.multiply(n1, n2);
8.     System.out.println("\n\nResult : " + n1 + " * " + n2 + " =
    " + result);
9. }
10. }
```